

|                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                   |  |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------|
| <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Supplementary Examination – Summer 2024</b><br><b>Course: B. Tech.                      Branch: Common to all Branches                      Semester : III</b><br><b>Subject Name &amp; Code: Engineering Mathematics – III (BTBS301/BTES301)</b><br><b>Max Marks: 60                      Date:29/06/2024                      Duration: 3 Hrs.</b> |                                                                                                                                                                                   |  |           |
| <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. Use of non-programmable scientific calculators is allowed.<br>3. Assume suitable data wherever necessary and mention it clearly.                                                                                                                                                                                                       |                                                                                                                                                                                   |  |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                   |  | Marks     |
| <b>Q. 1</b>                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Solve Any Two of the following.</b>                                                                                                                                            |  | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                                                        | Find the Laplace transform of $\frac{\sin 2t}{t}$ .                                                                                                                               |  | 6         |
| B)                                                                                                                                                                                                                                                                                                                                                                                                                        | Find the Laplace transform of $\int_0^t \left( \frac{e^{-at} - e^{-bt}}{t} \right) dt$ .                                                                                          |  | 6         |
| C)                                                                                                                                                                                                                                                                                                                                                                                                                        | Find the Laplace transform of $\operatorname{erf}(\sqrt{t})$ .                                                                                                                    |  | 6         |
|                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                   |  |           |
| <b>Q.2</b>                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Solve Any Two of the following:</b>                                                                                                                                            |  | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                                                        | Find the inverse Laplace transform of $\log\left(1 + \frac{1}{s^2}\right)$                                                                                                        |  | 6         |
| B)                                                                                                                                                                                                                                                                                                                                                                                                                        | Using Partial Fraction method, find the inverse Laplace Transform $\frac{s}{(s^2+1)(s^2+4)}$                                                                                      |  | 6         |
| C)                                                                                                                                                                                                                                                                                                                                                                                                                        | Find the inverse Laplace transform of $\frac{4s+15}{16s^2-25}$                                                                                                                    |  | 6         |
|                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                   |  |           |
| <b>Q. 3</b>                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Solve any Two of the following:</b>                                                                                                                                            |  | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                                                        | Find the Fourier transform of $f(x) = \begin{cases} 1, & \text{for }  x  < 1 \\ 0, & \text{for }  x  > 1 \end{cases}$ . Hence evaluate that $\int_0^\infty \frac{\sin x}{x} dx$ . |  | 6         |
| B)                                                                                                                                                                                                                                                                                                                                                                                                                        | Find the Fourier sine transform of $e^{- x }$ , and hence show that $\int_0^\infty \frac{x \sin mx}{1+x^2} dx = \frac{\pi e^{-m}}{2}$ , $m > 0$ .                                 |  | 6         |
| C)                                                                                                                                                                                                                                                                                                                                                                                                                        | Evaluate the integral $\int_0^\infty \frac{dx}{(a^2+x^2)(b^2+x^2)}$ .                                                                                                             |  | 6         |
|                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                   |  |           |
| <b>Q.4</b>                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Solve any Two of the following:</b>                                                                                                                                            |  | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                                                        | Form the partial differential equation by eliminating the arbitrary function from $z = f(x^2 - y^2)$ .                                                                            |  | 6         |
| B)                                                                                                                                                                                                                                                                                                                                                                                                                        | The partial differential equations by eliminating the arbitrary constant $z = (x^2 + a)(y^2 + b)$                                                                                 |  | 6         |
| C)                                                                                                                                                                                                                                                                                                                                                                                                                        | Solve the following partial differential equations $p + 3q = 5z + \tan(y - 3x)$ where the symbols have got their usual meanings.                                                  |  | 6         |
|                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                   |  |           |
| <b>Q. 5</b>                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Solve any Two of the following:</b>                                                                                                                                            |  | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                                                        | Show that $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is a harmonic function and hence determine the corre-                                                                               |  | 6         |

|           |                                                                                                                     |          |
|-----------|---------------------------------------------------------------------------------------------------------------------|----------|
|           | sponding analytic function                                                                                          |          |
| <b>B)</b> | Evaluate $\oint_C \frac{e^{-z}}{z+1} dz$ where $C$ is the circle $ z  = 2$ and $ z  = \frac{1}{2}$                  | <b>6</b> |
| <b>C)</b> | Use Cauchy's integral formula to evaluate $\oint_C \frac{e^{2z}}{(z+1)^4} dz$ , where $C$ the circle is $ z  = 2$ . | <b>6</b> |
|           | <b>***END***</b>                                                                                                    |          |